

QUICK START AND SAMPLE ACQUISITION GUIDE

NovoCyt[®] Opteon Spectral Flow Cytometer

STARTUP

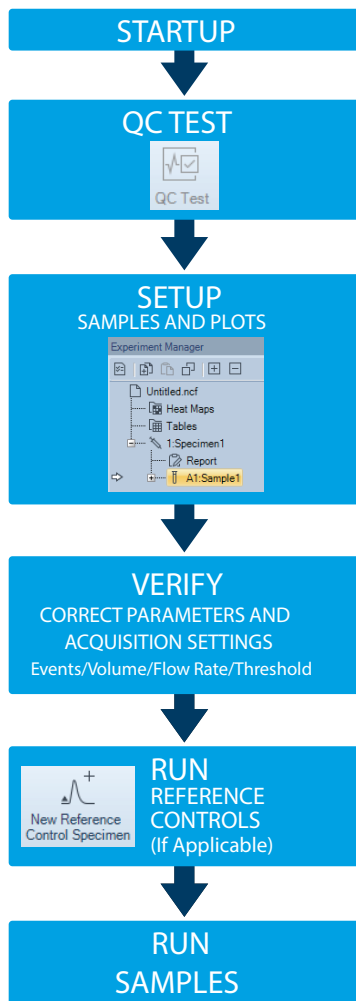
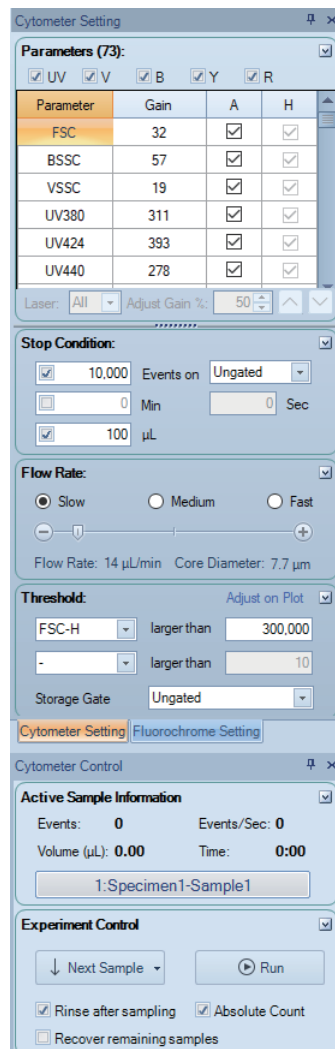
1. Check for adequate levels of the NovoFlow, NovoRinse, and NovoClean in the fluidic containers. Ensure that each is at least 1/2 full, and preferably empty waste container.
2. Press the power button on the front panel of the flow cytometer to turn on the instrument. Launch the NovoExpress (Opteon) software on the computer. Startup is automated and takes about 6.5 minutes. Check the instrument status as indicated by the LED indicator on the front panel:
Green = Normal.
The instrument is ready to use.
Orange = Warning.
Click on the status bar in the NovoExpress (Opteon) software to review the warning message.
Red = Error.
Click on the status bar to review the error message.
Refer to [Troubleshooting](#) in the associated instrument operator's guide.

QC TEST

1. Add 2 drops of the QC particles to 1 mL of dilution buffer (0.8 mL TE (0.2 µm filtered) and 0.2 mL NovoRinse (MilliQ Water) solution).
2. On the main menu, click on Instrument → QC Test and follow the software prompts to complete the test.
3. If the QC Test results display Pass or Acceptable, the instrument may be used to acquire data. If any QC Test results display Failed, refer to [Troubleshooting](#) in the associated instrument operator's guide.

FLUIDICS MANAGEMENT

1. The software will display a warning message instructing the user to add more reagents when the volumes of NovoFlow, NovoClean, and NovoRinse are less than the set thresholds of 1.25 L, 150 mL, and 150 mL, respectively.
2. To add reagents:
 - 1) Make sure the instrument is Off or in the Ready status.
 - 2) Disconnect the tubing from the container.
 - 3) Bring the container to the sink. Unscrew the cap and add more reagent to the container.
 - 4) Screw the cap back on, reconnect the quick coupler, and return the container back to the fluidics station. Ensure that the tubing is not twisted or kinked.
 - 5) If the instrument was in the Ready status, click Instrument → Fluidics Maintenance → Priming in the software to prime the system before running samples.
3. Similarly, the software will warn the user to empty the waste container when the waste volume exceeds 5.6 L.
 - 1) Follow the steps above to disconnect the waste container from the instrument. Dispose of the contents according to local safety guidelines.
 - 2) Add 600 mL of bleach to the waste container and reconnect the container to the instrument.



Cleaning & SHUTDOWN

Clean the SIP (Sample Injection Probe)

1. Click Instrument → Shut Down on NovoExpress software.
2. Select Clean Sample Injection Probe option, click Yes
3. Remove the sample tube holder.
4. Sample injection probe will move downward
5. Use a kimwipe soaked in NovoClean and gently clean the SIP
6. Replace the sample tube holder.
7. Place a tube of NovoRinse (MilliQ) in A1 and click OK
8. Instrument shut down automatically after clicking OK (~13 min)
- 9.

QUICK MAINTENANCE GUIDE

NovoCyte Opteon Spectral Flow Cytometer

PREVENTIVE MONTHLY MAINTENANCE

Clean the SIP (Sample Injection Probe)

1. Click Instrument → Shut Down on NovoExpress software.
2. Select Clean Sample Injection Probe option and then click Yes button in the prompted window.
3. Follow the instruction in the software to remove the sample tube from the tube holder. Sample injection probe will move downward after clicking the OK button.
4. Prepare a cotton swab or a damp soft cloth with NovoClean solution (1X) and gently wipe the outer surface of the sample injection probe. Click OK button after cleaning.
5. Place a tube of Milli-Q or 0.2 µm filtered DI water or NovoRinse solution (1X) in the tube holder. The instrument will shut down automatically after clicking OK button.

Clean the SIP Cleaning Apparatus

1. Power off the instrument.
2. Prepare a cotton swab soaked with NovoClean solution (1X). Wipe the SIP cleaning apparatus' lower surface back and forth until it is clean.
3. Prepare a damp cloth soaked with Milli-Q or 0.2 µm filtered DI water. Wipe the SIP cleaning apparatus back and forth at least five times.
4. Power on the instrument and start NovoExpress software. After the instrument is Ready, click Instrument → Fluidics Maintenance → Cleaning to finish.

Run the Priming/Debubble Procedure

1. Make sure the instrument is in Ready status.
2. Click Instrument → Fluidics Maintenance → Priming (or Debubble) to run the Priming or Debubble.

REPLACE FLUIDICS SYSTEM CONSUMABLES

1. The NovoFlow sheath fluid in-line filter is to be replaced every 6 months or when prompted by the NovoExpress software, whichever comes first. All other filters (waste and inlet filters, see below) should be replaced by following the prompts from the NovoExpress software when the maximum accumulated run time has been reached or click Instrument → Replace Fluidic System Consumables. You can see how much time has been accumulated since last filter change by clicking Instrument → Information. The notification will appear when 528 hours of instrument use has been reached.
2. Replace the NovoFlow sheath fluid in-line filter as described in associated instrument operator's guide or follow the notification prompt to change all the filters. Please refer to table below to order the proper fluidics maintenance kits with each filters included.

Fluidics Maintenance Kit	Filters included	Catalog Number
NovoCyte Opteon Spectral Flow Cytometer 1-year Fluidics Maintenance Kit	NovoFlow Sheath Fluid In-line Filter (2 pcs)	S2811-67120
	NovoFlow Sheath Fluid Inlet Filter (2 pcs)	
	NovoClean Solution Inlet Filter (2 pcs)	
	NovoRinse Solution Inlet Filter (2 pcs)	
	Waste Filter (2 pcs)	
NovoCyte Opteon Spectral Flow Cytometer 6-month Fluidics Maintenance Kit	NovoFlow Sheath Fluid In-line Filter (1 pcs)	S2811-67121
	NovoFlow Sheath Fluid Inlet Filter (1 pcs)	
	NovoClean Solution Inlet Filter (1 pcs)	
	NovoRinse Solution Inlet Filter (1 pcs)	
	Waste Filter (1 pcs)	

